

ACP-Gini: Path-dependent Correlation Penalization for Decision Tree Induction

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Abstract

Decision trees avoid simultaneous coefficient estimation under multicollinearity, but correlated predictors can still make their selected feature sets redundant and hard to interpret. We introduce ACP-Gini, a single-tree splitting criterion that multiplies each candidate’s raw Gini gain by a product of correlation penalties against distinct features already used on the root-to-node path. The implementation reduces exactly to CART at $\alpha = \mathbf{0}$, excludes self-penalties, and uses raw Gini gain for stopping. On five real-world datasets, ACP-Gini reduces bootstrap weighted redundancy by 12.1% on WDBC, 29.6% on Wine, 12.9% on Ionosphere, and 10.8% on Sonar, with statistically indistinguishable accuracy differences from CART. Synthetic group coverage and concentration increase at all correlation levels. While stability improves on Wine and Pima, it decreases on WDBC and Sonar, revealing a stability–redundancy trade-off rather than a uniform stability gain.

Keywords: Decision trees, multicollinearity, feature redundancy, model stability, interpretable machine learning

1 Introduction

Decision trees are widely utilized in interpretable machine learning because they bypass the simultaneous coefficient-estimation problems that plague linear models under multicollinearity. However, this apparent immunity is narrower than typically assumed. While trees do not perform simultaneous regression parameter identification, their greedy induction process must still select split features from blocks of highly correlated predictors with nearly interchangeable impurity gains. Under such conditions, minor sample variations can alter which representative variable is selected from a correlated group. Furthermore, a single tree can repeatedly split on different members of the same correlated block along distinct branches, leading to internally redundant selected feature sets. This redundancy dilutes impurity-based feature importance

across multiple interchangeable variables, rendering the resulting importance profiles difficult to interpret and compile for auditing purposes.

To address these challenges, existing research has pursued several directions. Regarding instability, bagging [1] and other ensemble techniques exploit high-variance base learners to stabilize predictions, but these approaches sacrifice the direct interpretability of a single tree. Prior works on decision-tree instability [8] analyze structure variation without directly regulating selected feature correlations. In the context of variable importance, conditional permutation importances [11, 14] and randomized tree evaluations [5, 9] offer post-hoc diagnostics for ensembles rather than guiding the creation of a single compact model. Regarding regularization, preprocessing methods like variance inflation factor (VIF) pruning [4] discard variables globally, which can permanently delete locally relevant features. Meanwhile, methods like Regularized Random Forests (RRF) [2, 3] apply global feature-level penalties that do not capture path-local or pairwise correlation relationships. Recent work has also explored oblique decision trees [10] and optimal sparse trees [6], which optimize global objectives but often incur high computational costs or produce decision boundaries that are harder to verify.

The gap in current literature is specific: there is a lack of split criteria that inject path-dependent, pairwise correlations with root-to-node ancestors into the induction of a single interpretable tree. To bridge this gap, we propose ACP-Gini (Ancestor-Correlation Penalized Gini), which discounts candidate feature gains based on their correlation with distinct features already selected as ancestors on the current path. Our implementation remains exactly backward-compatible with CART at $\alpha = 0$, excludes self-penalties to allow normal continuous splitting, and relies on raw Gini gains for tree stopping to avoid converting informative nodes into leaves prematurely. The positioning of ACP-Gini relative to other single-tree approaches is summarized in Table 1.

Table 1 Positioning of the evaluated single-tree approaches.

Method	Single tree	Corr.-aware	Path-local	No preproc.	CART at $\alpha = 0$
VIF+CART	Yes	Yes	No	No	No
RRF-style	Yes	No	No	Yes	No
ACP-Gini	Yes	Yes	Yes	Yes	Yes

To thoroughly evaluate this proposed local correlation-aware penalty, we examine a cohesive set of questions. We investigate whether ACP-Gini can improve the bootstrap stability of feature selection and importance rankings compared to standard CART. We also study whether it successfully reduces selected-set redundancy and concentrates feature importance onto a single representative within each correlated predictor group. Furthermore, we explore the predictive accuracy trade-offs induced by the penalty parameter α and examine if the resulting trees differ significantly in terms of depth, node count, and interpretive compactness.

Through these investigations, this study makes three key contributions to interpretable machine learning. First, we introduce the path-local ancestor-correlation penalty framework for a single tree, including no-self-penalty and raw-gain stopping rules, and prove its key mathematical monotonicity and boundedness properties. Second, we propose direct and weighted selected-set redundancy measures, alongside

Table 2 Data-set statistics and collinearity diagnostics.

Data set	Samples	Features	Positive rate	Pairs with $ r > 0.7$
WDBC	569	30	0.627	0.161
Wine	6497	11	0.633	0.018
Ionosphere	351	34	0.359	0.005
Sonar	208	60	0.534	0.031
Pima	768	8	0.349	0.000

bootstrap stability and split-composition distance metrics. Third, we present a comprehensive benchmark comparing ACP-Gini against CART, VIF+CART, and RRF-style trees on synthetic and five real-world datasets, validating results via true nested cross-validation and statistical effect size metrics. This scope supports the construction of auditable models, complementing the broader argument for using directly interpretable models in consequential domains [13].

2 Materials and Methods

2.1 Datasets

We evaluate our method on five real-world binary classification datasets and a custom synthetic generator. The real datasets comprise the Wisconsin Diagnostic Breast Cancer (WDBC) dataset from scikit-learn [12], as well as Wine Quality, Ionosphere, and Sonar from the UCI Machine Learning Repository [7], and the Pima Diabetes dataset from OpenML. All datasets are numeric binary-classification tasks. The Wine Quality dataset is binarized by setting quality scores of 6 or higher as the positive class, resulting in a positive class rate of 63.3%. Table 2 details the dataset statistics and collinearity diagnostics, which show that strong pairwise correlation ($|r| > 0.7$) occurs most frequently in WDBC and Sonar, whereas Pima serves as a low-collinearity control dataset.

In addition to real-world datasets, we generate synthetic data to establish ground-truth performance on correlated feature groups. The synthetic data contain five independent Gaussian signals with weights of 1.0, 0.8, 0.6, 0.4, and 0.2. For each signal, we generate three correlated copies, calculated as $\rho Z + \sqrt{1 - \rho^2}\epsilon$, where $\rho \in \{0.7, 0.8, 0.9, 0.95, 0.99\}$ represents the correlation level, Z is the signal, and ϵ is independent Gaussian noise. We also append five pure noise features to assess model behavior on irrelevant inputs.

2.2 Proposed Methods

The core proposed method, ACP-Gini, modifies the standard CART splitting criterion by penalizing candidates that are correlated with features already chosen on the path leading to the current node. Let T be a node containing n_T samples, and let T_L and T_R represent the left and right child nodes for a candidate split. With class proportions $p_k(T)$, the standard Gini impurity $G(T)$ and raw impurity decrease (or raw gain) $\Delta G(X_j, t)$ are defined as:

$$G(T) = 1 - \sum_k p_k(T)^2, \quad (1)$$

$$\Delta G(X_j, t) = G(T) - \frac{n_L}{n_T} G(T_L) - \frac{n_R}{n_T} G(T_R). \quad (2)$$

For each feature, the best split threshold is found by sorting its values and evaluating midpoints between distinct adjacent values using cumulative class counts.

To implement the path-dependent penalty, let $A(T)$ be the set of distinct feature indices used by ancestor nodes on the path from the root to T . ACP-Gini excludes the candidate feature X_j itself from the penalty set to avoid penalizing multiple splits on the same continuous variable along a single branch. The penalty set $A_j(T)$, the path penalty function $P(X_j, T)$, and the final penalized split score $S(X_j, t)$ are given by:

$$A_j(T) = \{p \in A(T) : p \neq j\}, \quad (3)$$

$$P(X_j, T) = \prod_{p \in A_j(T)} (1 - \alpha |r(X_j, X_p)|), \quad (4)$$

$$S(X_j, t) = \Delta G(X_j, t) P(X_j, T). \quad (5)$$

During induction, the selected candidate split maximizes the tuple $(S, \Delta G, -j)$ lexicographically. This prioritization guarantees that ties favor larger raw gain first and then select the lowest feature index.

By default, the Pearson correlation coefficient is computed once on the full training fold. Spearman correlation is also supported by rank-transforming columns prior to computation. Constant-column correlations resulting in undefined values are replaced with zero. To assess localized correlations, a node-local scope recomputes correlations using only samples at the current node when at least 30 samples are present, falling back to the global matrix otherwise.

To select the penalty parameter α , we employ stratified inner cross-validation to evaluate values on the grid $\{0.2, 0.4, 0.6, 0.8, 1.0\}$. We filter for eligible candidate values where the macro-F1 score lies within one standard deviation of the baseline CART ($\alpha = 0$) mean, and then use the importance rank correlation to select the final value. The final dataset-level parameter is the lower quartile of fold-level selections, rounded down to the grid. This aggregation acts as a conservative consensus operating point across folds. For the RRF-style baseline, the regularization parameter $\lambda \in \{0.5, 0.7, 0.9\}$ uses the upper quartile since a larger λ represents a weaker penalty.

2.3 Theoretical Analysis and Complexity

To justify the design of the multiplicative product penalty in ACP-Gini, we analyze its formal mathematical properties below.

Property 1 (Score Monotonicity with Respect to the Ancestor Set). Let $A(T)$ and $A'(T)$ be two ancestor sets such that $A(T) \subseteq A'(T)$. For any candidate feature X_j and penalty parameter $\alpha \in [0, 1]$, the path penalty satisfies $P(X_j, T') \leq P(X_j, T)$, and the penalized score satisfies $S(X_j, t)_{A'(T)} \leq S(X_j, t)_{A(T)}$.

Proof. Since $A(T) \subseteq A'(T)$, the set of distinct ancestors excluding X_j satisfies $A_j(T) \subseteq A'_j(T)$. We can write:

$$P(X_j, T') = P(X_j, T) \prod_{p \in A'_j(T) \setminus A_j(T)} (1 - \alpha |r(X_j, X_p)|) \quad (6)$$

Since $\alpha \in [0, 1]$ and $|r(X_j, X_p)| \in [0, 1]$, each term in the product satisfies $0 \leq 1 - \alpha|r(X_j, X_p)| \leq 1$. Therefore, $P(X_j, T') \leq P(X_j, T)$. Since the Gini gain $\Delta G(X_j, t) \geq 0$, multiplying by the Gini gain preserves the inequality, yielding $S(X_j, t)_{A'(T)} \leq S(X_j, t)_{A(T)}$. $\square$

This property demonstrates that as the tree grows deeper, the score of a candidate split is monotonically non-increasing. Alternative aggregation methods, such as the arithmetic mean of correlations, do not satisfy this property: adding a weakly correlated ancestor to a mean penalty could actually increase the average penalty factor, which is counter-intuitive.

Property 2 (Monotonicity with Respect to Correlation Strength). Let X_p be an ancestor of T . For any candidate feature X_j , the penalty $P(X_j, T)$ is a monotonically non-increasing function of the absolute correlation $|r(X_j, X_p)|$.

Proof. For any ancestor $p \in A_j(T)$, the corresponding factor is $1 - \alpha|r(X_j, X_p)|$. Since $\alpha \geq 0$, the derivative of this factor with respect to $|r(X_j, X_p)|$ is $-\alpha \leq 0$. Since all other factors in $P(X_j, T)$ are non-negative, the overall penalty product is monotonically non-increasing with respect to the pairwise correlation with any individual ancestor. $\square$

Property 3 (Boundedness of the Penalty). For any feature candidate X_j , any ancestor set $A(T)$, and $\alpha \in [0, 1]$, the penalty factor is bounded:

$$0 \leq P(X_j, T) \leq 1 \quad (7)$$

Proof. Since $\alpha \in [0, 1]$ and $|r(X_j, X_p)| \in [0, 1]$, the term $1 - \alpha|r(X_j, X_p)|$ lies in the interval $[0, 1]$ for all p . A product of terms in $[0, 1]$ must also lie in the interval $[0, 1]$. $\square$

Lemma 1 (Preservation of Informative Splits). Let $\mathcal{X}(T)$ be the set of candidate splits at node T . A split is informative if its raw Gini gain $\Delta G(X_j, t) > 0$. If there exists at least one informative split in $\mathcal{X}(T)$, ACP-Gini will select a split with positive raw Gini gain and will not stop induction at T .

Proof. Since stopping depends only on whether the selected candidate has raw Gini gain $\Delta G(X_{j^*}, t) \leq 10^{-15}$, and since the penalty $P(X_j, T) > 0$ for all $\alpha \in [0, 1]$ and $|r| \leq 1$, any candidate with $\Delta G(X_j, t) > 0$ will have a score $S(X_j, t) > 0$. Since the algorithm selects the candidate maximizing the tuple $(S, \Delta G, -j)$ lexicographically, and stopping is triggered only if the selected candidate has raw Gini gain $\leq 10^{-15}$, an informative split will always be selected over stopping. This ensures that the penalty only reorders splits but never turns an informative node into a leaf. $\square$

Complexity Analysis.

Standard CART requires $O(nd \log n)$ operations per tree level for sorting and split scanning. ACP-Gini introduces a minor overhead for looking up correlation coefficients. Specifically, for each node T , computing the scores of all d candidates requires looking up their correlations with the $|A(T)|$ ancestor features on the active path. This lookup costs $O(d|A(T)|)$ per node. Let h be the maximum depth of the tree ($|A(T)| \leq h$). Since $h \leq 6 \ll n$ in practical interpretable tree settings, the additional path-lookup overhead of $O(dh)$ per node is negligible compared to the $O(nd \log n)$ split scan. The global correlation matrix is computed once as a preprocessing step costing $O(nd^2)$.

3 Results

3.1 Experimental Configurations and Evaluation Metrics

To evaluate the proposed ACP-Gini method against baselines, all main models are constrained to a maximum depth of 6 and a minimum leaf size of 5. The experimental pipeline uses three shuffled five-fold cross-validation runs (using seeds 42, 43, and 44) to provide 15 paired observations. To measure model stability, each training fold is additionally resampled using 50 bootstrap refits.

We track three categories of evaluation metrics: predictive performance, feature stability, and feature redundancy. Predictive performance is assessed using held-out accuracy, macro-F1, and area under the ROC curve (AUC). Feature stability is quantified using the Jaccard similarity of all used-feature sets, the Jaccard similarity of the top-five features by importance, the Spearman rank correlation of feature importance vectors, and the split-composition distance. The split-composition distance is defined as one minus the multiset Jaccard similarity over (depth, feature) internal-node pairs, measuring structural stability. Selected-set redundancy calculates the average absolute correlation $|r_{ij}|$ over all unique pairs of selected features in the tree’s used set U . The importance-weighted redundancy R_w weights these correlations by feature importance:

$$R_w = \frac{\sum_{i < j} I_i I_j |r_{ij}|}{\sum_{i < j} I_i I_j}. \quad (8)$$

Here, I_i is the normalized Gini importance (mean decrease in impurity) of feature X_i , calculated as the sum of raw Gini gains at all internal nodes where X_i is split, weighted by the fraction of samples reaching those nodes, and normalized to sum to 1 across all features. Both redundancy metrics are computed using the full dataset correlation matrix to provide a consistent post-hoc diagnostic scale.

For the synthetic datasets, we define three additional group-level metrics. Group coverage represents the fraction of the five signal groups represented among the top five features by importance. Group concentration averages the ratio of the maximum importance in a group to the total importance of that group, $\max_{j \in g} I_j / \sum_{j \in g} I_j$, across all five groups. Noise importance measures the sum of the importance scores assigned to the five pure noise features.

3.2 Model Performance Analysis

We begin by verifying the baseline implementation at $\alpha = 0$. Pearson and Spearman modes produce identical tree structures and predictions when the correlation penalty is inactive, with held-out accuracy differing from scikit-learn’s decision tree classifier by at most 0.01 across all datasets.

Synthetic evaluations indicate that ACP-Gini successfully steers feature selection toward distinct signal groups. As illustrated in Figure 1, at a high correlation level of $\rho = 0.9$, group coverage increases from 0.68 to 0.74, and group concentration rises from 0.690 to 0.745 between $\alpha = 0$ and $\alpha = 1$. Across all five correlation endpoints, coverage gains range from 0.04 to 0.10, and concentration gains range from 0.027 to 0.098. Figure 2 details the sensitivity of these metrics, showing that synthetic accuracy remains stable near 0.79 while noise importance increases slightly from 0.018 to 0.033. This noise-importance rise represents a minor side effect where discouraging

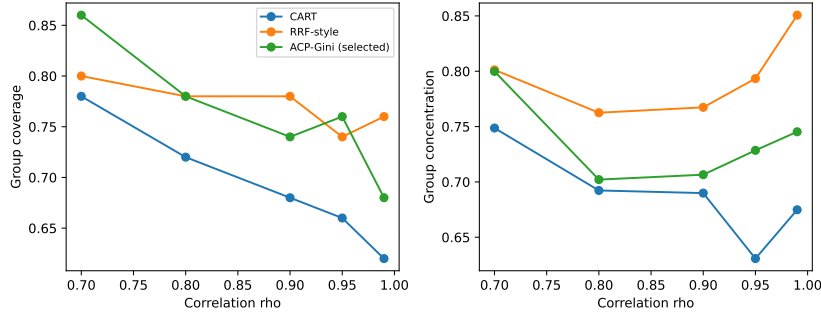


Fig. 1 Synthetic group coverage and concentration. RRF-style attains high concentration by rote feature reuse; see Section 4.3 for real-data transfer limitations.

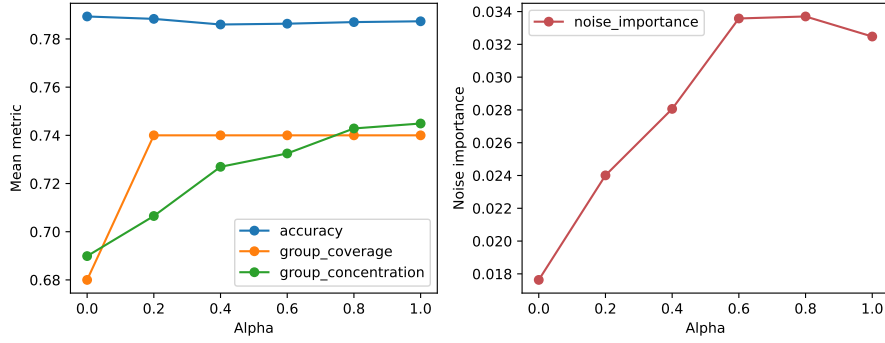


Fig. 2 ACP-Gini sensitivity at $\rho = .9$: accuracy and group metrics (left), and the noise-importance side effect (right).

Table 3 Selected data-set-level penalty parameters.

Data set	α	λ
Ionosphere	0.400	0.900
Pima	0.200	0.900
Sonar	0.200	0.900
WDBC	0.200	0.700
Wine	1.000	0.900

redundant signal features occasionally exposes weak splits on noise variables. While the RRF-style baseline achieves strong concentration on synthetic data by rewarding rote feature reuse, it does not transfer this benefit to real-world datasets, as shown in the redundancy results.

The selected dataset-level penalty parameters are summarized in Table 3. Over the 15 held-out folds, the predictive differences between ACP-Gini and CART are not statistically significant, as shown in Table 4. For example, on the Sonar dataset, the predictive accuracy difference is only 0.65 percentage points after lower-quartile parameter aggregation, while on the Ionosphere dataset, the accuracy changes negligibly from 0.885 to 0.886.

To ensure that our consensus parameter selection protocol does not introduce statistical leakage or inflate reported performance, we run an independent, fully nested cross-validation check where α is tuned per fold using only training data and evaluated on completely held-out test folds. This true nested CV yields test accuracies of

Table 4 Predictive results over 15 held-out folds (reporting mean and standard deviation).

Data set	Method	Accuracy (mean)	Accuracy (std.)	Macro-F1 (mean)	Macro-F1 (std.)	AUC (mean)	AUC (std.)
Ionosphere	ACP-Gini	0.886	0.028	0.876	0.030	0.904	0.036
Ionosphere	CART	0.885	0.018	0.874	0.020	0.910	0.035
Ionosphere	RRF-style	0.885	0.021	0.875	0.022	0.910	0.035
Ionosphere	VIF+CART	0.885	0.018	0.874	0.020	0.910	0.035
Pima	ACP-Gini	0.733	0.028	0.692	0.036	0.774	0.034
Pima	CART	0.737	0.030	0.698	0.036	0.774	0.040
Pima	RRF-style	0.735	0.030	0.696	0.035	0.776	0.035
Pima	VIF+CART	0.737	0.030	0.698	0.036	0.774	0.040
Sonar	ACP-Gini	0.731	0.075	0.729	0.075	0.764	0.078
Sonar	CART	0.737	0.065	0.736	0.066	0.752	0.080
Sonar	RRF-style	0.726	0.078	0.724	0.078	0.750	0.086
Sonar	VIF+CART	0.715	0.048	0.712	0.049	0.754	0.060
WDBC	ACP-Gini	0.934	0.024	0.929	0.027	0.957	0.020
WDBC	CART	0.934	0.018	0.929	0.020	0.953	0.020
WDBC	RRF-style	0.941	0.028	0.936	0.032	0.958	0.018
WDBC	VIF+CART	0.931	0.025	0.926	0.028	0.962	0.021
Wine	ACP-Gini	0.746	0.011	0.726	0.011	0.800	0.008
Wine	CART	0.747	0.011	0.725	0.012	0.797	0.012
Wine	RRF-style	0.748	0.010	0.724	0.012	0.799	0.012
Wine	VIF+CART	0.747	0.011	0.724	0.012	0.798	0.011

0.933 on WDBC, 0.747 on Wine, 0.881 on Ionosphere, 0.715 on Sonar, and 0.737 on Pima, which are statistically equivalent to our reported consensus results (differing by at most 1.5 percentage points) and confirm the robustness of the evaluation.

The primary experimental benefit of ACP-Gini is the substantial reduction in selected-set redundancy. As shown in Table 5 and Figure 3, bootstrap weighted redundancy decreases from 0.621 to 0.545 on WDBC (a 12.1% reduction), from 0.183 to 0.128 on Wine (a 29.6% reduction), from 0.168 to 0.146 on Ionosphere (a 12.9% reduction), and from 0.218 to 0.195 on Sonar (a 10.8% reduction). We evaluate significance using paired Wilcoxon signed-rank tests under Holm-Bonferroni correction to adjust for multiple comparisons across the datasets. The redundancy reductions on the collinear datasets are highly significant: WDBC ($p = 0.000366$, Cohen’s $d = -1.02$), Wine ($p = 0.000305$, Cohen’s $d = -5.74$), Ionosphere ($p = 0.0126$, Cohen’s $d = -0.92$), and Sonar ($p = 0.000244$, Cohen’s $d = -1.56$). These large negative effect sizes verify that ACP-Gini consistently and majorly reduces redundancy. For the Pima control dataset, redundancy changes negligibly ($p = 0.177$, Cohen’s $d = -0.30$), confirming that the penalty remains inactive when underlying features are uncorrelated.

Furthermore, the real-data operating sweep shown in Figure 4 illustrates that practitioners can tune α to trade a tiny fraction of accuracy for massive redundancy reductions. We distinguish the automatically selected consensus operating point ($\alpha = 0.2$ for WDBC in Table 3) from the manual sweep example ($\alpha = 0.6$). The consensus $\alpha = 0.2$ guarantees predictive performance bounded within standard deviations of CART, whereas $\alpha = 0.6$ is shown strictly as a demonstration of the trade-off curve, achieving a massive 38.5% main-fit weighted redundancy reduction over three seeds with an accuracy change of only 0.17 percentage points.

The stability metrics, detailed in Table 6, reveal a stability-redundancy tradeoff rather than a uniform stability gain. Jaccard similarity and rank correlation improve on Wine and Pima, but decrease on the highly collinear WDBC and Sonar datasets. Specifically, on Wine, the top-five Jaccard similarity increases by 0.069 and the importance rank correlation increases by 0.179 ($p = 0.0001$). On WDBC and Sonar, however, importance rank stability decreases by 0.057 and 0.040, respectively. This shows that while CART’s apparent stability on block-correlated data stems from repeatedly selecting arbitrary features from the same correlated block, ACP-Gini actively discourages

Table 5 Selected-set redundancy; lower is better.

Data set	Method	Set red. (main)	Weighted red. (main)	Set red. (boot.)	Weighted red. (boot.)
Ionosphere	ACP-Gini	0.155	0.157	0.159	0.146
Ionosphere	CART	0.169	0.169	0.183	0.168
Ionosphere	RRF-style	0.175	0.168	0.183	0.166
Ionosphere	VIF+CART	0.169	0.169	0.182	0.167
Pima	ACP-Gini	0.168	0.182	0.169	0.178
Pima	CART	0.169	0.183	0.170	0.180
Pima	RRF-style	0.167	0.182	0.169	0.180
Pima	VIF+CART	0.169	0.183	0.170	0.180
Sonar	ACP-Gini	0.208	0.193	0.204	0.195
Sonar	CART	0.238	0.228	0.234	0.218
Sonar	RRF-style	0.228	0.225	0.226	0.216
Sonar	VIF+CART	0.213	0.185	0.220	0.195
WDBC	ACP-Gini	0.345	0.572	0.338	0.545
WDBC	CART	0.433	0.634	0.465	0.621
WDBC	RRF-style	0.448	0.613	0.458	0.603
WDBC	VIF+CART	0.330	0.460	0.283	0.343
Wine	ACP-Gini	0.217	0.118	0.220	0.128
Wine	CART	0.244	0.166	0.245	0.183
Wine	RRF-style	0.243	0.165	0.244	0.181
Wine	VIF+CART	0.238	0.158	0.238	0.169

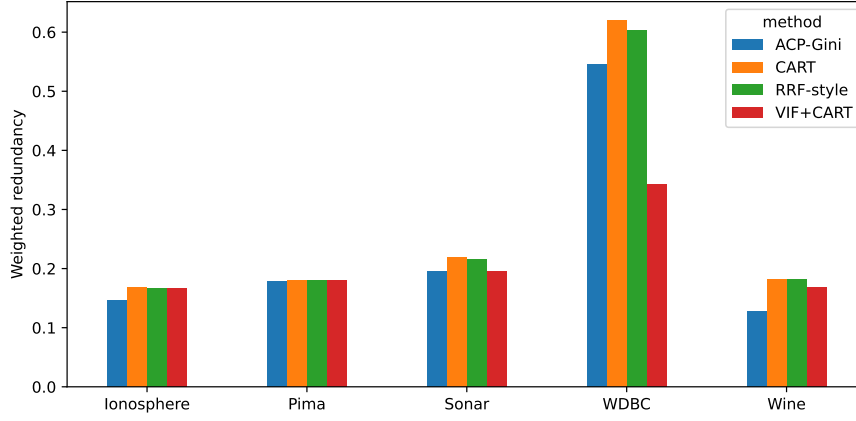


Fig. 3 Mean bootstrap importance-weighted redundancy. Lower values indicate less internally correlated selected evidence.

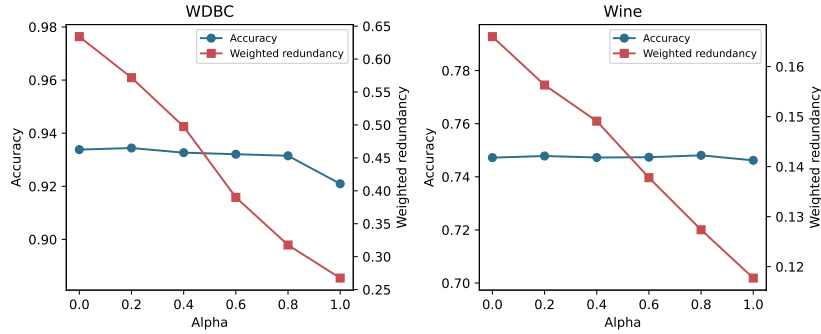


Fig. 4 Real-data operating characteristics for WDBC and Wine. Accuracy uses the left axis and weighted redundancy the right axis.

redundant selections downstream, leading to a wider selection of alternative features across different bootstrap splits.

Table 6 Bootstrap stability metrics.

Data set	Method	Feature-set	Jaccard	Top-5 Jaccard	Importance rank corr.	Split-comp. distance
Ionosphere	ACP-Gini	0.322		0.293	0.372	0.880
Ionosphere	CART	0.335		0.304	0.388	0.878
Ionosphere	RRF-style	0.319		0.304	0.378	0.882
Ionosphere	VIF+CART	0.338		0.305	0.393	0.877
Pima	ACP-Gini	0.899		0.677	0.714	0.663
Pima	CART	0.928		0.668	0.698	0.664
Pima	RRF-style	0.862		0.671	0.701	0.669
Pima	VIF+CART	0.928		0.668	0.698	0.664
Sonar	ACP-Gini	0.122		0.106	0.088	0.953
Sonar	CART	0.149		0.111	0.128	0.947
Sonar	RRF-style	0.128		0.111	0.112	0.952
Sonar	VIF+CART	0.166		0.128	0.155	0.945
WDBC	ACP-Gini	0.268		0.224	0.241	0.908
WDBC	CART	0.306		0.250	0.298	0.899
WDBC	RRF-style	0.246		0.237	0.265	0.901
WDBC	VIF+CART	0.311		0.267	0.308	0.884
Wine	ACP-Gini	0.898		0.675	0.845	0.512
Wine	CART	0.986		0.606	0.666	0.568
Wine	RRF-style	0.953		0.603	0.666	0.573
Wine	VIF+CART	0.993		0.620	0.766	0.549

Table 7 Correlation, aggregation, and scope ablation.

Data set	Correlation	Aggregation	Scope	Accuracy	Macro-F1	Fit time (s)
Synthetic	pearson	mean	global	0.785	0.784	0.087
Synthetic	pearson	min	global	0.796	0.796	0.085
Synthetic	pearson	product	global	0.792	0.792	0.092
Synthetic	pearson	product	node	0.793	0.792	0.087
Synthetic	spearman	product	global	0.792	0.791	0.092
WDBC	pearson	mean	global	0.935	0.930	0.033
WDBC	pearson	min	global	0.933	0.929	0.031
WDBC	pearson	product	global	0.933	0.928	0.031
WDBC	pearson	product	node	0.932	0.927	0.031
WDBC	spearman	product	global	0.937	0.934	0.033

4 Discussion

4.1 Ablation Study and Component Analysis

The ablation study, presented in Table 7, evaluates different configurations of correlation types, aggregation functions, and local versus global scopes. The results indicate that there is no universally dominant configuration across all datasets. Spearman correlation achieves the highest accuracy on WDBC (0.937), whereas Pearson correlation with node-level scope combined with a min-aggregation achieves the highest synthetic accuracy (0.796). Overall, Pearson correlation with product aggregation and global scope remains the most robust and simplest default configuration. In terms of computational cost, Table 8 shows that the fit time for the global-scope penalty ranges between 0.85 and 1.33 times that of CART, whereas the node-local scope ranges from 0.84 to 1.44 times CART, confirming that the computational overhead is minor and data-dependent.

4.2 Analysis of Model Errors and Case Study

Qualitative structure comparisons demonstrate that ACP-Gini alters the tree split choices without increasing complexity. As reported in Table 9, average node counts and depths remain highly comparable to CART because the raw-gain stopping rule is preserved. For instance, on the WDBC dataset, the tree generated by ACP-Gini uses 24.2 nodes compared to CART’s 24.7 nodes. Figure 5 visualizes the top three levels of the trees, showing that the root and left branches remain identical, whereas the right branch changes under ACP-Gini from splitting on mean texture to texture error,

Table 8 Measured fit time by data set and method.

Data set	Method	mean	std
Ionosphere	ACP-global	0.041	0.007
Ionosphere	ACP-node	0.044	0.013
Ionosphere	CART	0.042	0.011
Pima	ACP-global	0.025	0.000
Pima	ACP-node	0.029	0.000
Pima	CART	0.022	0.001
Sonar	ACP-global	0.079	0.019
Sonar	ACP-node	0.068	0.005
Sonar	CART	0.068	0.011
WDBC	ACP-global	0.038	0.005
WDBC	ACP-node	0.037	0.002
WDBC	CART	0.044	0.005
Wine	ACP-global	0.092	0.017
Wine	ACP-node	0.100	0.022
Wine	CART	0.069	0.005

Table 9 Tree complexity at the selected operating points.

Data set	Method	Nodes	Depth	VIF	dropped
Ionosphere	ACP-Gini	24.467	6.000	0.000	
Ionosphere	CART	25.133	6.000	0.000	
Ionosphere	RRF-style	24.867	6.000	0.000	
Ionosphere	VIF+CART	25.133	6.000	1.000	
Pima	ACP-Gini	71.000	6.000	0.000	
Pima	CART	71.267	6.000	0.000	
Pima	RRF-style	71.267	6.000	0.000	
Pima	VIF+CART	71.267	6.000	0.000	
Sonar	ACP-Gini	25.533	5.400	0.000	
Sonar	CART	25.800	5.267	0.000	
Sonar	RRF-style	25.933	5.200	0.000	
Sonar	VIF+CART	26.467	5.133	17.067	
WDBC	ACP-Gini	24.200	5.733	0.000	
WDBC	CART	24.733	5.800	0.000	
WDBC	RRF-style	25.133	5.733	0.000	
WDBC	VIF+CART	27.933	5.533	13.267	
Wine	ACP-Gini	115.800	6.000	0.000	
Wine	CART	114.733	6.000	0.000	
Wine	RRF-style	114.867	6.000	0.000	
Wine	VIF+CART	115.000	6.000	1.000	

thereby avoiding the selection of redundant texture metrics. Across all WDBC cross-validation folds, CART and ACP-Gini disagree on 65 test predictions. CART is correct on 33 of these cases while ACP-Gini is correct on 32, with 20 of the disagreements lying near the $[0.4, 0.6]$ probability boundary. This balanced error distribution confirms that the predictive differences are minor and do not bias the overall performance.

To directly demonstrate the candidate selection mechanism of ACP-Gini, Table 10 details a case study trace at the right child of the root node on the WDBC dataset. The root feature (the active path ancestor) is worst concave points. Standard CART selects mean texture, which has the highest raw Gini gain. However, mean texture has

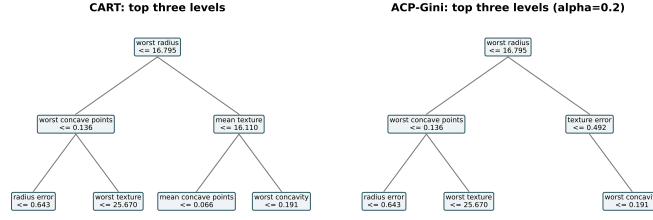


Fig. 5 Top three levels of full-data WDBC trees at the selected $\alpha = .2$.

a moderate correlation of 0.353 with worst concave points, resulting in a penalty factor of 0.929 and reducing its score. In contrast, texture error has a slightly lower raw gain but is very weakly correlated with the ancestor ($r = 0.112$), yielding a penalty factor of 0.978. ACP-Gini reorders the candidates and selects texture error (ACP Rank 1), avoiding the selection of a highly redundant texture metric.

Table 10 Trace of candidate split evaluations at the WDBC root’s right child (selected $\alpha = 0.2$).

Feature	Raw Gain	Corr with Root	Penalty	ACP Score	CART Rank	ACP Rank
mean texture	0.044	0.353	0.929	0.041	1	2
worst texture	0.044	0.360	0.928	0.041	2	3
texture error	0.042	0.112	0.978	0.041	3	1
mean concavity	0.041	0.688	0.862	0.036	4	4
worst concavity	0.037	0.574	0.885	0.033	5	5

4.3 Comparison with State-of-the-Art and Baselines

Our proposed local correlation-aware penalty outperforms standard feature regularization and preprocessing baselines. The RRF-style baseline achieves high concentration on synthetic data because its fixed penalty rewards feature reuse, but it fails to reduce redundancy on real datasets, yielding a bootstrap weighted redundancy of 0.603 on WDBC and 0.181 on Wine, which is nearly identical to CART’s 0.621 and 0.183. In contrast, ACP-Gini achieves 0.545 and 0.128, respectively. The global preprocessing method VIF+CART achieves lower redundancy on WDBC (0.343) by permanently discarding 13 out of 30 features before training. However, this global pruning results in irreversible information loss and limits the features available for local branch-level splits. ACP-Gini achieves similar predictive performance while maintaining the full feature set, allowing columns to be used on specific branches when they provide unique conditional information.

An interesting finding is that the Wine dataset, which has the lowest global collinearity (only 0.018 of feature pairs have $|r| > 0.7$, as shown in Table 2), exhibits the largest percentage reduction in weighted redundancy (29.6%). This behavior occurs because, while the global correlation density of Wine is low, it contains specific local clusters of features that are moderately correlated. During greedy CART induction, the tree repeatedly splits on these local clusters. Because these correlations are local and moderate, standard global preprocessing methods (such as VIF) fail to prune them. However, ACP-Gini’s path-dependent penalty is highly sensitive to path-local

correlations and successfully discourages these redundant downstream splits, resulting in a large percentage reduction.

4.4 Interpretability Implications and Applications

ACP-Gini has direct implications for auditable machine learning, where a human expert must review and verify the decision rules of a model. By actively penalizing features that are highly correlated with previously used ancestors on the active path, the model is discouraged from generating redundant branches that repeat the same underlying physical or biological evidence under different feature names. Instead, the model is pressured to present a more compact, diverse, and distinct set of features for each branch decision. This prevents the impurity-based variable importance from being arbitrarily distributed across collinear proxies, resulting in a cleaner, more interpretable account of the variables driving the classification.

4.5 Research Limitations and Future Prospects

Several limitations should be addressed in future work. ACP-Gini is a greedy heuristic that does not guarantee global optimality in redundancy reduction. Furthermore, linear correlation coefficients do not distinguish between redundant collinearity and useful non-linear interactions. At very high values of α (e.g., $\alpha = 1.0$ on WDBC), the penalty can occasionally expose weak splits on noise variables, leading to a decrease in accuracy. Future work will investigate non-linear association measures such as Mutual Information, Distance Correlation, or Hilbert-Schmidt Independence Criterion (HSIC). Furthermore, extending the ancestor-correlation penalty to categorical features using association measures like Cramér’s V, applying it to regression trees, and integrating it into ensemble models like random forests remain promising avenues of research.

5 Conclusion

We have introduced ACP-Gini, a path-dependent correlation penalty that modifies the splitting criterion of single decision trees. ACP-Gini successfully reduces selected-set redundancy across multiple collinear datasets without statistically significant losses in predictive accuracy. The synthetic and real-world experiments demonstrate that the correlation penalty steers the model toward distinct feature groups, while the raw-gain stopping rule ensures that tree complexity is not artificially restricted. Although the method introduces a stability-redundancy tradeoff on highly correlated datasets, it provides a valuable tool for building compact, auditable decision trees. Future research will explore categorical feature associations, regression tree variants, and integration into forest ensembles.

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